A Strategy for Sizing and Optimizing the Energy System on Long-Range AUVs

Ariel Chiche, Göran Lindbergh, Ivan Stenius, and Carina Lagergren

***Abstract*—Autonomous underwater vehicles (AUVs) are utilized in a variety of unmanned missions, such as environmental surveys or guarding coastal waters. Currently, the most commonly used energy storage is a secondary battery pack. A potential solution for increasing the range even further on larger AUVs is to utilize hybrid fuel cell/battery systems. Fuel cell systems can have a significantly increased specific energy compared to batteries and are now starting to become well-developed technologies. However, in an underwater environment, both hydrogen and oxygen need to be stored for the fuel cell. This study considers the sizing of a hybrid fuel cell/battery system with respect to operational power profiles, i.e., how to select the best combination of fuel cell stack, battery pack, and reactant storage, and compares its volume and weight with conventional battery systems. This article presents a sizing strategy developed for this purpose, where the optimum combination of fuel cell output power and battery size is analyzed. The strategy is implemented on power profiles from real AUV missions. The sizing strategy will outline the techniques to reach the optimum hybrid configuration capable of meeting the power demand at any time of the mission. Results show that for longer missions, the volume of the hybrid system becomes significantly lower than for the battery system, meaning that the endurance can be increased thanks to the use of fuel cells.**

***Index Terms*—Autonomous underwater vehicle (AUV), battery, fuel cell (FC), hydrogen storage, optimization.**

I. Introduction

AUTONOMOUS underwater vehicles (AUVs) are typically used for missions, such as ocean monitoring and sampling of water or oceanographic data collection. Well-developed sensors, sonars, and underwater communication systems make these AUVs very efficient and capable of ambitious missions. However, all the onboard equipment requires power, and that is achieved at the expense of endurance of the AUV. Therefore, the power required for increasing range and endurance is a key performance parameter when evaluating an AUV.

Manuscript received December 13, 2019; revised November 13, 2020 and February 12, 2021; accepted February 17, 2021. Date of publication April 2, 2021; date of current version October 13, 2021. This work was supported by the Swedish Foundation for Strategic Research (SSF) through the Swedish Maritime Robotic Centre (SMaRC IRC 15-0046). This work was presented in part at the IEEE OES-AUV 2018 Symposium, Porto, Portugal, Nov. 6–9, 2018. *(Corresponding author: Ariel Chiche.)*

Associate Editor: N. Cruz.

Ariel Chiche, Göran Lindbergh, and Carina Lagergren are with the Division of Applied Electrochemistry, School of Engineering Sciences in Chemistry, Biotechnology and Health, KTH Royal Institute of Technology, 114 28 Stockholm, Sweden (e-mail: chiche@kth.se; gnli@kth.se; carinal@kth.se).

Ivan Stenius is with the Division of Naval Architecture, School of Engineering Sciences, KTH Royal Institute of Technology, 114 28 Stockholm, Sweden (e-mail: stenius@kth.se).

Digital Object Identifier 10.1109/JOE.2021.3062047

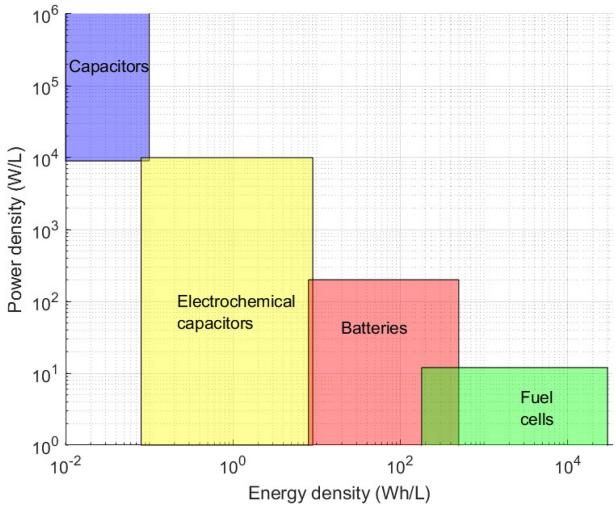


Fig. 1. Ragone plot (based on [14]) comparing the power density and energy density of different types of energy-storage systems.

The Swedish Maritime Robotics Centre (SMaRC) aims to design and develop AUVs to achieve missions in unknown waters and perform environmental analyses. These AUVs will need not only highly advanced sensors and sonars for the studies, but also an efficient power system. The research in SMaRC focuses on three target application areas: 1) ocean production of algae; 2) safeguarding society; and 3) environmental monitoring, including missions in unexplored polar marine environments. The general objective is to increase the autonomy and endurance of marine robots to reduce the reliance on ships and crews, resulting in a reduction of the costs of these operations. Today, AUVs are usually powered by secondary (rechargeable) batteries, such as Li-ion batteries [1]–[4]. These have a high efficiency but both weight and volume increase rapidly with the energy needs. Moreover, working in a cold environment has a negative influence on the battery performance, since that decreases when the temperature drops, which is likely to occur during many underwater missions [5]. Compared to batteries, fuel cell (FC) systems can offer a higher energy density in watthour per liter but have a lower power density in watt per liter, as shown in the Ragone plot; see Fig. 1. Such a plot is often used to compare the power density and energy density of different types of energy storage systems and it highlights the potential of FC systems in terms of energy density. This means that FCs can provide larger endurance for the AUV although the maximum power is lower. Several studies in which AUVs powered by a pure FC system

This work is licensed under a Creative Commons Attribution 4.0 License. For more information, see https://creativecommons.org/licenses/by/4.0/



Fig. 2. AUV "HUGIN" used to run test missions within SMaRC.

or a hybrid one (FC/batteries) are presented in the literature [1], [6]–[8]. These studies show the feasibility of and interest in investigating such systems for AUVs. Hybridization aims to use FCs for the basic power demands of the AUV. When power demands are high, the FCs are supplemented by batteries, which can then be recharged when power demands are low [9], [10].

Although there are several established examples of hybrid FC/battery vehicles such as cars [11] and buses [12], there is as yet no standard drive cycle for AUVs. Additionally, the inherent underwater nature of AUVs generates additional constraints, e.g., neutral buoyancy [13], not faced by land-based vehicles. Furthermore, there is no regenerative braking that can be used to charge the battery on an AUV. The battery can only be charged by the FC stack.

This study presents an analysis to select the best combination of FC, battery, and reactants storage (hereafter: hybrid system) with regard to volume and weight for use in AUVs. The study considers aspects such as FC stack, hydrogen storage, oxygen storage, battery pack, battery state-of-charge (SoC), compensation needed to reach neutral buoyancy, and operational power profile. A sizing strategy for designing hybrid FC/battery systems and comparing volume and weight of a full battery system and a hybrid system is developed and used on real power profiles from AUV missions. In a preliminary study [15], only the weight and volume of the battery pack, the FC, and the reactants were estimated. This article extends the study by also considering the battery SoC during the mission, the total buoyancy of the energy system, and the compensation needed to reach neutral buoyancy. This article also introduces an optimization strategy to choose the output FC power and keep the system as compact as possible. By using this sizing strategy, the main goal is to identify the energy system that is most volumetric and weight efficient for a given mission. The methodology was applied both to missions undertaken by an existing AUV, Kongsberg's HUGIN (see Fig. 2), and to hypothetical missions.

II. FCs for AUVs

A. FC System

There are several types of FCs, but the proton exchange membrane FC (PEMFC) is the most suitable for powering AUVs, since the operating temperature is low (15 °C–80 °C) compared to other FCs [16]. The PEMFC is currently the most common FC type, and the technology is relatively mature and robust. Interest in the use of PEMFC, for example, propulsion in passenger cars, has led to strong growth in the FC industry. A PEMFC consumes oxygen and hydrogen to produce electricity, water, and heat according to the following electrochemical reactions:

$$\text{Anode} : \mathrm{H_2} = 2\mathrm{H^+} + 2e^- \tag{1}$$

$$\text{Cathode} : \frac{1}{2}\mathrm{O_2} + 2\mathrm{H^+} + 2e^- = \mathrm{H_2O} \tag{2}$$

$$\text{Total} : \frac{1}{2}\mathrm{O_2} + \mathrm{H_2} = \mathrm{H_2O}. \tag{3}$$

The FC system usually requires storage of hydrogen, for which several alternatives exist. Given that this study concerns operations underwater, oxygen storage is also needed, and this is less commonly explored than the hydrogen storage. The choice of the hydrogen and oxygen storage for the FC system has a very strong impact on the total weight and volume of the system.

B. FC-Powered AUVs

Since endurance is one of the main challenges for the performance of AUVs, primary and secondary batteries , in some projects over the past two decades, have been replaced by FCs [1], [6], [8]. The projects have had different goals and ranges, but they all use FC-based systems to power the AUV and have been successfully conducted. Even if there is no commercial FC-powered AUV as yet, these examples show that this solution is credible and likely to be developed in the coming years. The Japanese URASHIMA project [1], in which a PEM FC and Li-ion batteries power the AUV, and a metal hydride is used as fuel storage, started in 1998. In 2005, a 56-h mission was performed, travelling 317 km and reaching a depth of 800 m, and in 2009, a 1000-h test was performed with an energy efficiency of 60%. Other projects such as IDEFIX [1] (France) and [7], which use compressed hydrogen and compressed oxygen for its hybrid system, and DEEPC (Germany) [8], which achieved a 60-h endurance mission in 2004, point to strong and continuing interest and success in using FCs for AUVs. Table I provides an overview of projects found in the open literature using this kind of technology to power AUVs. All the presented projects showed promising capabilities. However, there is a need to define a methodology to design efficient AUVs and take advantage of both the FC and the battery [17].

III. Design Philosophy

A. Methodology

Several methodologies for optimizing the energy system on AUVs have been developed in recent years. One important step is to define the boundary conditions [20]. Such conditions highlight the key parameters that influence the design of the AUV and what needs to be optimized. In space exploration, it is mass [21]. For underwater vehicles, it is typically volume and neutral buoyancy [17]. In this article, the results will, therefore, primarily be compared regarding volume. Since the key is to minimize both volume and weight, where volume is more important than weight, each component of the system will be evaluated according to its technical performance and

TABLE I
EXAMPLES AND CHARACTERISTICS OF FC-POWERED (FROM AUVS [1], [3], [6], [8], [18], AND [19])

Name	Type of system	Characteristics
Urashima I Japan 1998	PEM + Li-ion battery Metal hydride for H_2	10 tons - 10 m length-1.4 m diameter 300 km cruise 3500 m depth 2003: 26 km in 6 h 2004: 220 km in 44 h 2005: 317 km in 56 h
Urashima II Japan 2009	PEM + Li-ion battery	600 h endurance 3000 km cruise 2009: 1000 h test 2013: 180 m depth mission
DeepC Germany 2003	1.5 kW PEM + Li-ion battery H_2 and O_2 storage: Gas at 250 and 350 bar	2 tons 2 m/s 60 h 4000 m depth
IDEFIX France 2009	PEM + Li-ion battery H_2 300 bar 50L O_2 250 bar 50L	1.6 tons 6 m length 400 m depth 300 km range
Taiwan 2013	1 kW PEM + Lead-acid battery Pure O_2 fed	491 kg 1.05 knots

to its volume. The remaining available space of the AUV can be used for the implementation of other systems such as new sensors and sonars, but only if the weight is reduced as well. It is important to note that the size of the tubing is not considered in the study, although it is mentioned in Section V. The general sizing procedure described in this section is summarized in Algorithm 1.

The hybrid system is composed of the FC system and the battery pack. It is assumed that the FC is maintained at constant power during the entire mission [22]. The battery pack is being charged by the FC when the power demand of the AUV is below the FC output power and discharged when the power demand is higher than the FC power. The assumption of constant FC power is simplistic but is commonly used when sizing hybrid systems [9], [22]–[25]. Furthermore, using dimensionless parametric characterization of the mission is beneficial in the design of hybrid AUV systems, since the hybridization of an FC and electrical energy storage unit depends on the system power requirements. Several dimensionless parameters can be used to characterize the system [9]. The degree of hybridization with respect to power ($\mathrm{DoH_{Power}}$) is the most important one; it compares the power provided by the FC, P_{fc} and the maximum power demand P_{max}

$$\mathrm{DoH_{Power}} = \frac{P_{\mathrm{max}} - P_{\mathrm{fc}}}{P_{\mathrm{max}}} \tag{4}$$

If $\mathrm{DoH_{Power}}$ is high (close to 1), it indicates a high power output from the batteries compared to the FCs, meaning that the batteries will provide a greater part of power during power peak demand. For land-based vehicles, the degree of hybridization will be chosen to optimize for regenerative braking that charges the battery [26]. For underwater vehicles, the battery can only be charged by the FC stack, and hence, the degree of hybridization is not the same as for wheel-based land vehicles. Moreover, the power profile of a mission is also an important input when designing the energy storage system. It dictates the energy consumption and the dimensionless parameters, e.g., the power ratio P and the time ratio T

$$P = \frac{P_{\mathrm{max}}}{P_{\mathrm{base}}} \tag{5}$$

$$T = \frac{t_1}{t_2}. \tag{6}$$

For periodic behavior, the power profile can be divided into cycles, where each of them corresponds to a battery discharging time t_1 and charging time t_2. The average power P_{average} can also be estimated

$$P_{\mathrm{average}} = \frac{\int \mathrm{Power}(t)dt}{\mathrm{Time}} \tag{7}$$

thanks to the mission's power profile where "Power" is the power demand and "Time" is duration of the mission.

Examples of power profiles are shown in Figs. 3, 6, and 10. Since the FC is supposed to charge the battery during base power demand, it is important to highlight which phase of the operation is limited: the charging or discharging phase of the battery. Parameters T and P determine the criterion for which this phenomenon is limited. The charging is limited if

$$\mathrm{DoH_{Power}} < \frac{2 \times T \times (P-1)}{P \times (T+1)^2}. \tag{8}$$

When the charge is limited, the battery cannot be fully charged by the FC during the base power demand of the AUV. If the discharge is limited, the battery is not fully discharged after peak power utilization. Using these parameters, the sizing strategy has been applied to different cases and power profiles. The main idea is to estimate the total energy need of the mission and, thus, to determine the size of the FC, the amount of hydrogen and oxygen needed, and the size of the battery. This gives the total volume and weight of the hybrid system for three oxygen and nine hydrogen storage alternatives using their energy density and specific energy and compares it with full primary and secondary battery systems. For the batteries, an efficiency of 95% and a utilization of 80% of the total SoC window are considered. The batteries are assumed to be fully charged at the beginning of the mission. The overall FC system efficiency η_{fc} is assumed to be 60% in all the following estimations, as in [25].

B. Hydrogen and Oxygen Storage

Storage of hydrogen and oxygen represents a real issue for underwater applications, especially the storage of oxidant. In most other FC applications, the oxidant is the ambient air. For commercial PEMFC cars, the hydrogen is stored as compressed gas at 350 or 700 bar. Table II compares several solutions for storage of both hydrogen and oxidant, whereas Table III shows the gravimetric and volumetric densities. The content of

TABLE II
GAS STORAGE CHARACTERISTICS (FROM [1], [8], AND [20])

Storage	Advantages	Disadvantages
Gaseous H_2	Simpler Cheaper No pre-processing Up to 700 bar	Heavy storage High pressure H_2 embrittlement Explosive hazards if leaking
Liquid H_2	Very little H_2 wasted High energy density	Low temperature High complexity Losses during refuelling process Safety issues Explosive hazards if leaking
Hydrocarbons	Cheap Simple reaction Extensively used to produce H_2 Low pressure	High complexity Emission of CO CO impurities: danger for FC Sensitive to water, and carbon Decrease of storage efficiency
Metal hydrides	High H_2 content Simple process	Need to apply heat Heavy system
Chemical O_2	Spontaneous reaction Common chemical	High cost High weight Highly oxidizing chemicals used
Compressed O_2	Simple mechanism Up to 690 bar	High cost
Cryogenic liquid oxygen (LOX)	High specific energy	Complex High cost

hydrogen per volume and weight of storage are compared. The storage includes the tank, its wall, and insulation.

To estimate the necessary amounts of hydrogen and oxygen, the FC is assumed to run at constant power during the entire mission. The electric energy delivered by the FC (E_{fc}) is then

$$E_{\text{fc}} = P_{\text{fc}} \times \text{Time} \tag{9}$$

with "Time" being the mission duration. The amount of hydrogen needed m_{H_2} is estimated using the low heating value of hydrogen LHV_{H_2} and the overall FC system efficiency η_{fc}

$$m_{\text{H}_2} = \frac{E_{\text{fc}}}{\text{LHV}_{\text{H}_2} \times \eta_{\text{fc}}}. \tag{10}$$

LHV_{H_2} is used since the produced water is not entirely in the liquid form. The amount of oxygen consumed is estimated using stoichiometric (3) and (10)

$$m_{\text{O}_2} = \frac{m_{\text{H}_2} \times M_{\text{O}_2}}{M_{\text{H}_2} \times 2} \tag{11}$$

where M_{H_2} and M_{O_2} are the molar masses of hydrogen and oxygen, respectively. The weight and volume of the storage can then be estimated by dividing the gas weights simply calculated by the gravimetric and the volumetric density of each storage solution (see Table III).

C. Battery Design

Since hybrid systems are considered in this study, sizing the battery stack is part of the methodology. The battery SoC during the mission is a key parameter that needs to be taken into account when sizing the battery pack. Several articles about strategies for energy management explain how to estimate the SoC as a function of time [27], [28]. Being able to follow the SoC makes it possible to check if the proposed design is suitable and to study how it behaves.

As a starting point, the battery is sized to fulfill one cycle of peak power demand Es_{cycle}. The estimation takes the battery efficiency B_{eff} and its utilization BU into account

$$Es_{\text{cycle}} = \frac{(P_{\text{max}} - P_{\text{fc}}) \times t_1}{B_{\text{eff}} \times \text{BU}} \tag{12}$$

The battery SoC

$$\text{SoC}(t+1) = \text{SoC}(t) - \frac{\int(\text{Power}(t) - P_{\text{fc}})dt}{Es_{\text{cycle}}} \tag{13}$$

is also estimated for each time step of the mission (each second): The battery is charged when the power demand $Power$ is below the FC power. If that is the case and if the battery SoC reaches a set maximum value (around 1), the battery is no longer being charged.

The battery sizing is included in a recursive loop: If the SoC reaches a minimum value (0 or 0.1 for example) at some point, the size of battery pack Es_{cycle} is multiplied by a given coefficient (here 1.01) and the SoC as function of time is estimated again. This goes on until the SoC never reaches the set low limit. This procedure ensures a full energy system able to fulfill the mission with the smallest possible battery pack coupled with the FC. The final battery energy content is then $Es_{\text{cycle}_{\text{final}}}$. This procedure is also presented in Algorithm 1.

Furthermore, C_{rate} of the battery has to be considered. C_{rate} of 1 C means 1 h for a full charge (or discharge), 0.5 C or C/2 is a 2-h charge, whereas 2 C means 30 min. The C_{rate} can have an important impact on the battery's aging process [29]. If C_{rate} is higher than 1, the battery should be designed for power; if it is lower than 1, it should be designed for energy. This rate is estimated using the maximum battery power $P_{\text{max}_{\text{Bat}}}$ and its energy content

$$C_{\text{rate}} = \frac{P_{\text{max}_{\text{Batt}}}}{Es_{\text{cycle}_{\text{final}}}}. \tag{14}$$

On a practical level, it is better to have C_{rate} close to, or below, 1 so that the battery can deliver all its energy and limit the ageing process [29]–[31].

Eventually, when the battery is correctly dimensioned, a new dimensionless parameter can define the entire energy system: the degree of hybridization with respect to energy ($\text{DoH}_{\text{Energy}}$)

$$\text{DoH}_{\text{Energy}} = \frac{E_{\text{fc}} - Es_{\text{cycle}_{\text{final}}}}{E_{\text{fc}}}. \tag{15}$$

If $\text{DoH}_{\text{Energy}}$ is close to 1, it means that most of the energy comes from the FC, i.e., from the hydrogen and oxygen storage. In contrast, if it is close to 0, it means that the major part of the energy comes from the battery pack.

D. Optimization Strategy

One important step is to find the optimal FC output power, when the volume of the entire system is limited. The idea is to estimate the volume and weight of the hybrid system for several configurations (FC output power within a certain range) and find a local minimum. By changing the FC output power, the amount of hydrogen and oxygen to store is changing, as is the size of the

TABLE III
FUEL AND OXYGEN STORAGE CHARACTERISTICS (FROM [9])

Storage	Gravimetric density $kg_{H_2}/kg_{storage}$	Volumetric density $kg_{H_2}/m^3{}_{storage}$
Liquid H_2	0.142	70.80
200 bar H_2	0.012	16.00
700 bar H_2	0.060	35.00
Metal hydride	0.020	115.0
Liquid O_2 (LOX)	0.786	661.0
153 bar O_2	0.213	166.7
H_2O_2	0.303	348.8

battery pack. The result is a combination of an FC output power and a battery size that can fulfill the mission while minimizing volume and weight.

The FC size is estimated using linear interpolation for weight and volume from several commercial PEMFC stacks [32], [33]. The type of hydrogen and oxygen storage does not change the optimum point found. Indeed, changing the hydrogen and oxygen storage influences the weight and volume but the minimum will appear for the same FCs output power and, therefore, the same amount of hydrogen and oxygen needed. On a practical level, the optimization is made using a loop, which makes the full weight and volume estimation for each FCs power.

E. Neutral Buoyancy of the Energy System

Since this study concerns an underwater application, it is, thus, important to take into account the buoyancy of the system [13], [34]. Submarine subsystems typically aim to have a volumetric mass of the energy system ρ close to 1 kg/L, which means neutral buoyancy. If it is below the density of water, approximately 1 kg/L, some weight has to be added to increase the volumetric mass. Commonly, lead or steel with high density is used. If the volumetric mass on the other hand is above 1 kg/L, the buoyancy needs to be increased by adding some volume with a lighter material than water (most efficient gain in buoyancy is achieved by only filling that volume with air; however, typically this is achieved with some lightweight foam). Considering such compensation to ensure neutral buoyancy is crucial when designing an AUV and the volumetric mass ratio of each energy system is, therefore, estimated. In this article, it will be assumed that steel is used to increase the mass of the system and empty volume with only air inside will be used to decrease the mass of the system (referred to as steel or air compensation). In addition, the compensation (air or steel) needed to reach neutral buoyancy can be estimated by solving the following system of equations:

$$\begin{cases} \frac{W_{\text{hybrid-system}}+W_{\text{comp}}}{V_{\text{hybrid-system}}+V_{\text{comp}}} = 1 \\ \frac{W_{\text{comp}}}{V_{\text{comp}}} = \rho_{\text{comp}} \end{cases} \tag{16}$$

where $W_{\text{hybrid-system}}$ and $V_{\text{hybrid-system}}$ are the weight and volume of the hybrid system without compensation, respectively, W_{comp} and V_{comp} are the weight and volume of the compensation needed, respectively, and ρ_{comp} is the density of the considered compensation. If the buoyancy is above 1, the compensation is air, with a density of 1.23 g/L. If the buoyancy is below 1, the compensation is steel with a density of 8 kg/L. The neutral buoyancy procedure is also presented in Algorithm 1.

Algorithm 1: Sizing, Optimization and Neutral Buoyancy Procedure.

Input : Power profile
Output: Fuel cell size, battery size, hydrogen and oxygen needed, air or steel compensation
`load(`*Power profile*`)`;
for $P_{fc} \rightarrow min(Power)$ **to** $max(Power)$ **do**
  `Estimate(`*Dimensionless parameters*`)`;
  `Estimate(`*Fuel cell size*`)`;
  `Estimate(`*Hydrogen needed:* m_{H_2}`)`;
  `Estimate(`*Oxygen needed:* m_{O_2}`)`;
  `Estimate(`*Battery pack:* Es_{cycle}`)`;
  `Check(`*SoC at any time of the mission*`)`;
  while $SoC > 0.1$ **do**
    `Update(`$Es_{cycle} = Es_{cycle} \times 1.01$`)`;
  end
  `Update(`$Es_{cycle_{Final}} = Es_{cycle}$`)`;
  Size the hybrid system for each fuel cell power;
end
Selection of the fuel cell power leading to the smallest hybrid system;
`Estimate(`*Total volume and weight of the energy system for several reactants storage solutions*`)`;
`Estimate(`*Volumetric mass of the energy system,* ρ`)`;
if $\rho > 1$ **then**
  Compensation is air;
  `Estimate(`*Air volume and weight needed to reach neutral buoyancy*`)`;
if $\rho < 1$ **then**
  Compensation is steel;
  `Estimate(`*steel volume and weight needed to reach neutral buoyancy*`)`;
Volume-optimized energy system with neutral buoyancy;

IV. RESULTS

Several missions were conducted using HUGIN, an AUV equipped with Li-ion batteries, pictured in Fig. 2. In this article, two representative missions from the tests with HUGIN are used, one short (5-h mission), case 1, and one longer mission (40-h mission), case 2. All parameters such as t_1, t_2, P_{max}, P_{base}, P_{fc}, and mission duration mentioned earlier are estimated using these real power profiles. For nonperiodic profiles, t_1 and t_2 were estimated by taking the longest peak power demand and the longest low power time. Finally, a further case, case 3, is studied, exemplifying how arbitrary user-defined power profiles can be analyzed.

A. Case 1: Short-Range Mission

Case 1, which corresponds to the power profile from the short-mission test (see Fig. 3), was implemented in the sizing and optimization process (see Algorithm 1), and the optimal power output of 900 W was found minimum in the blue curve in Fig. 4. The average power demand is 929 W, as presented in Fig. 3.

In this case, using an FC power of 900 W, the battery charging and discharging times are found to be comparable, 20.2 and 29.6 min, respectively. $\text{DoH}_{\text{Power}}$ is low [0.31; see (4)], which means that most of the power is taken from the FC, whereas the obtained power ratio *P* is high [2.08; see (5)], showing an important difference between the base power and the maximum

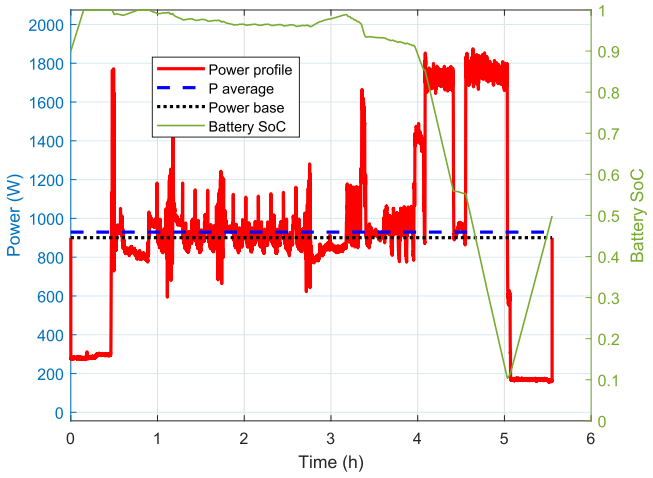


Fig. 3. Power profile for a short-range HUGIN mission with a 900-W FC stack and estimation of the battery SoC.

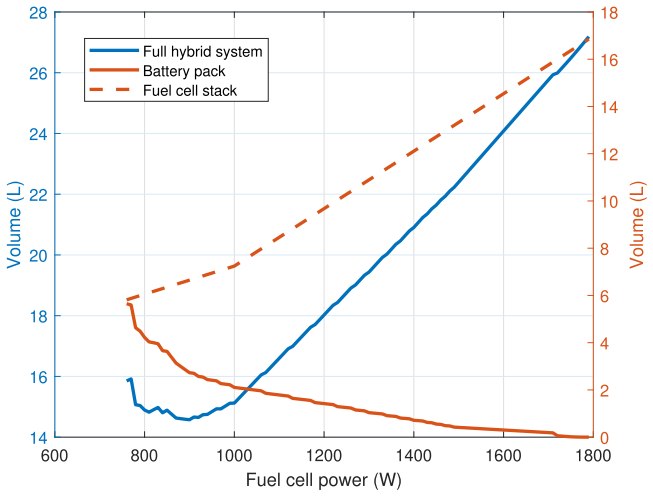


Fig. 4. Volume of the entire hybrid system (on the left axis) of the battery pack and the FC stack (on the right axis) as a function of the FC output power for the short HUGIN mission (liquid oxygen and metal hydride storage).

TABLE IV
BATTERY CHARACTERISTICS (FROM [35] AND [36])

Battery	Specific energy	Energy density	Density
Li-ion	217 Wh/kg	337 Wh/L	1.55 kg/L
Primary battery	480 Wh/kg	1080 Wh/L	2.25 kg/L

one. C_{rate} is equal to 1.07 [see (14)] meaning that the battery does not need to be designed for a specific purpose (energy or power) and can be implemented with this design. According to the criterion defined in (8), this means that battery charging is limited, not discharging. Furthermore, the battery SoC during the mission is estimated and it never goes below the minimum limit (0.1). $\text{DoH}_{\text{Energy}}$ is 0.82 [see (15)], which means that the energy primarily comes through the FC from the hydrogen and oxygen storage. It is interesting to note that there are two peaks of power demand at the end of the mission. The battery energy is, therefore, depleted down to 10% of SoC. The characteristics of the batteries considered in this article are listed in Table IV.

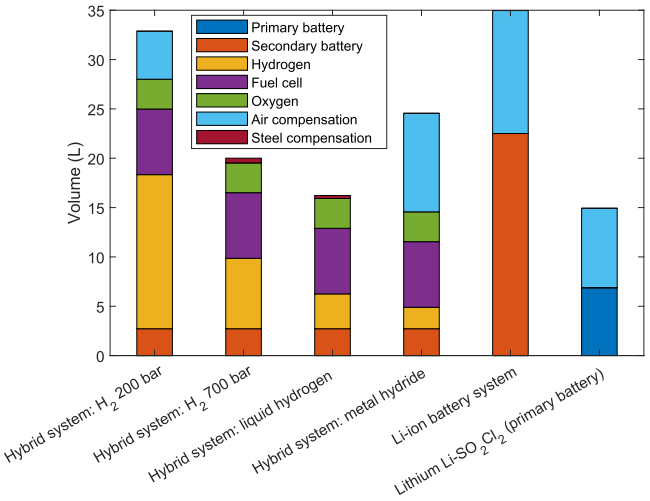


Fig. 5. Volume distribution for several hydrogen storage solutions and liquid oxygen storage using the power profile from the short HUGIN mission.

The short-mission power profile was used to simulate and compare the size of different energy systems; see Fig. 5. While varying the hydrogen storage for the hybrid systems, liquid oxygen storage was used in all cases. The results show that the hybrid system is more volume efficient than the secondary battery one, although the final volume is highly dependent on the method used to store hydrogen. In addition, the type of compensation, i.e., air or steel, and its size varies with the type of storage. It is also interesting to note that, for the two battery cases and the liquid oxygen combined with metal hydride system, the contribution of the air compensation to the total volume is very high. When a storage system for hydrogen with higher volumetric density is chosen, the volume of the hybrid system comes closer to the full primary battery. The FC itself contributes to a substantial part of the total volume in the hybrid systems. For a short mission, the difference between hybrid and pure battery systems is not significant but an interesting trend in favor of optimized hybrid systems appears.

B. Case 2: Long-Range Mission

A long-range 40-h mission (see Fig. 6), conducted by HUGIN was also implemented in the optimization process (see Algorithm 1). Fig. 7 shows that the best power output of the FC is 980 W, whereas Fig. 6 presents the average power demand of the mission 923 W. With an FC power of 980 W, the analysis of the power profile gave a low $\text{DoH}_{\text{Power}}$ [0.2; see (4)], which was expected since the power profile is relatively steady and peak power demands rarely occur.

The low $\text{DoH}_{\text{Power}}$ shows a high utilization of the FC in this case. Moreover, for this longer mission, the charging is limited, but it is interesting to note that the ratio T is close to 1, again, due to the stable power profile. In addition, the end of the mission requires less power than the rest of it; therefore, the battery SoC reaches 100%, meaning that the battery is no longer being charged when the power demand is low. It should lead to lower reactants consumption. The high C_{rate} [5.10; see (14)] highlights the need for power-designed batteries. Since

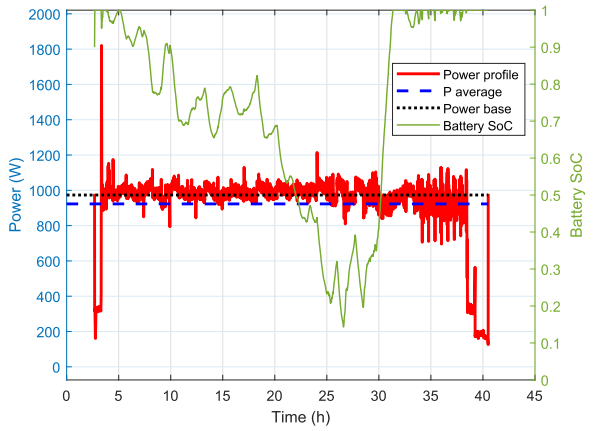


Fig. 6. Power profile for a long-range HUGIN mission with a 980-W FC stack and estimation of the battery SoC.

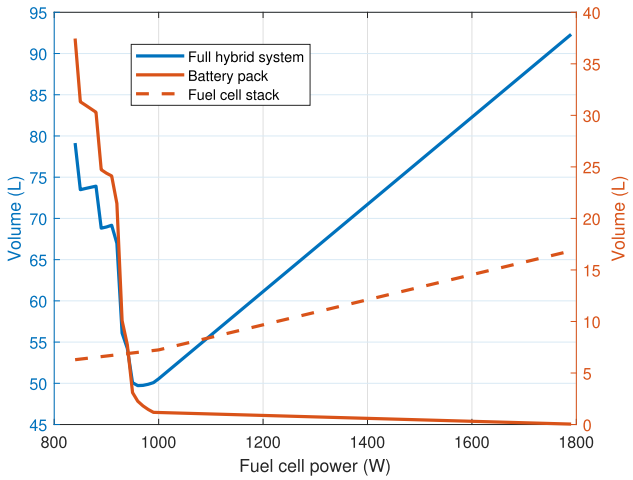


Fig. 7. Volume of the entire hybrid system (on the left axis) and of the battery pack and the FC stack (on the right axis) as a function of the FC output power for the long HUGIN mission with liquid oxygen and metal hydride storage.

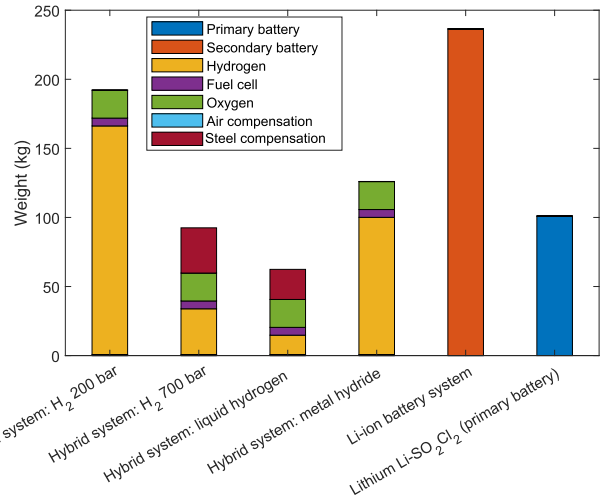


Fig. 8. Weight distribution for several hydrogen storage solutions and liquid oxygen storage using the power profile from the long HUGIN mission with neutral buoyancy.

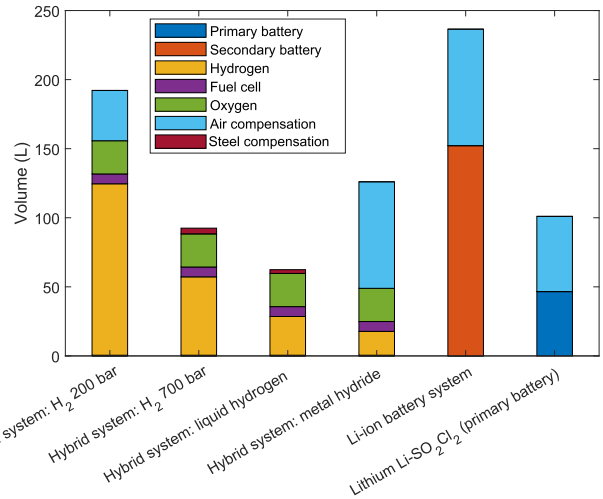


Fig. 9. Volume distribution for several hydrogen storage solutions and liquid oxygen storage using the power profile from the long HUGIN mission with neutral buoyancy.

the longest discharging time is not the interval that draws the most power, the battery SoC never reaches 0.1, meaning that the battery stack is slightly oversized. The very high $\text{DoH}_{\text{Energy}}$ [0.99; see (15)] is due to the very short peaks of high power demand and highlights an uneven distribution of energy storage in the system (liquid oxygen and metal hydride storage) between the FC system and the battery pack.

The weight and volume estimation for the 40-h mission, presented in Figs. 8 and 9, shows that the hybrid system is more weight and volume efficient than the secondary battery solution. In addition, for some combinations of hydrogen storage and liquid oxygen, e.g., 700 bar or liquid hydrogen, the hybrid system can also compete with the primary battery system in terms of weight and volume. In the case of an AUV, the metal hydride alternative can be a very interesting hydrogen storage solution. In that case, the oxygen storage becomes the most volume-demanding part. It is important to estimate the compensations needed. Storing hydrogen in liquid form or as a gas at 700 bar, combined with the liquid oxygen storage, has very low volumetric mass, resulting in the necessary steel addition to maintain a neutral buoyancy in the AUV. In contrast, the metal hydride and 200-bar hydrogen storage options have a high volumetric mass, which means that some air is needed in the system to ensure neutral buoyancy. When comparing the compensations needed for the short mission with the ones needed for the long mission, it appears that the compensations needed increase with the duration of the mission. The addition of air as compensation highly impacts upon the total volume of the system, whereas the addition of steel mostly increases the total weight. Once again, the compensations needed for the two battery cases represent an important part of the total volume of the system. When comparing the two HUGIN missions (long and short), it seems that when the mission length increases, the performance of the hybrid system, in terms of volume, comes closer to the full primary battery system. The two studied missions have similar power demands, but different lengths and behavior. The first analysis showed that when the mission duration increases, the

TABLE V
MISSION SPECIFICATIONS FOR THE HYPOTHETICAL POWER PROFILE

t_1	t_2	P_{max}	P_{base}	P_{fc}	$Time$
20 min	30 min	1800 W	900 W	1260	20 h

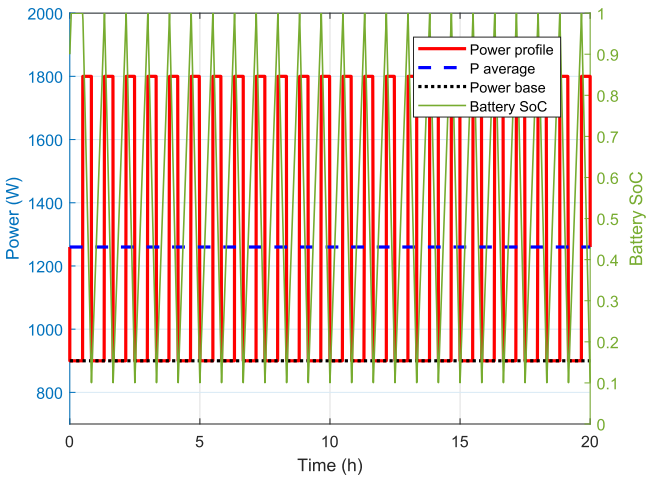


Fig. 10. Hypothetical power profile with a 1260-W FC stack and estimation of the battery SoC.

hybrid system becomes more relevant. The choice of hydrogen storage is shown to have a significant impact on the final weight and volume efficiency of the hybrid system. The battery in the hybrid system seems to be negligible compared to the rest of the system, meaning that an oversized battery pack is not a serious issue in that case.

C. Case 3: User-Defined Mission

Generated simple and cyclic power profiles can be used to conduct the same weight and volume analysis as presented earlier. Instead of reading a power profile, the user can implement the key parameters t_1, t_2, P_{max}, P_{base}, P_{fc}, and duration of the mission into the sizing strategy. Such a hypothetical profile for a 20-h mission was generated to match the parameters from the short mission with HUGIN and its specifications are listed in Table V. An optimization (see Algorithm 1) was also made for this power profile (see Fig. 10), and led to an FC output power equal to 1260 W, as shown in Fig. 11. The power profile with the optimized FCs output power and the corresponding battery behavior is presented in Fig. 10.

The resulting power profile is highly cyclical and reaches the maximum power at every cycle, giving a more energy-intense system than for the previously studied power profiles. The analysis of the dimensionless parameters gives a time ratio T of 0.67 [see (6)], i.e., times for charging and discharging are comparable. According to the criterion in (8), the time for charging the battery from the FC system is limited. The power ratio P is 2 [see (5)], showing that the difference between the base and peak power demand is significant. Furthermore, in this hypothetical case, C_{rate} is high, 2.7 [see (14)], meaning that the battery pack has to be power optimized. This is due to the numerous cycles in the power profile and the relatively long duration of the mission. $\text{DoH}_{\text{Energy}}$, in this case, is 0.99 [see (15)], which shows an uneven

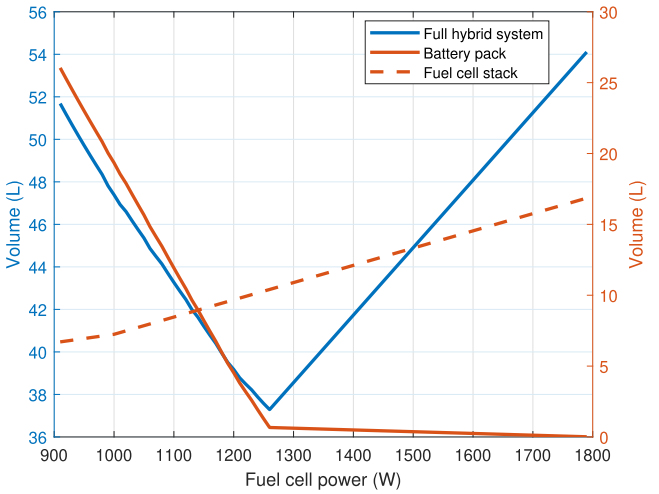


Fig. 11. Volume of the entire hybrid system (on the left axis) and of the battery pack and the FC stack (on the right axis) as a function of the FC output power for the hypothetical power profile with liquid oxygen and metal hydride storage.

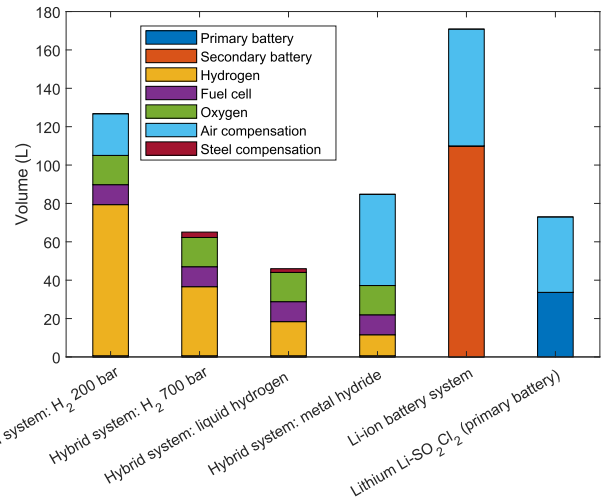


Fig. 12. Volume distribution for the hypothetical 20-h mission when using liquid oxygen and different hydrogen storage solutions with neutral buoyancy.

distribution in terms of energy storage between the battery pack and the FC system (hydrogen and oxygen stored).

By using the sizing strategy (see Algorithm 1) with input from Table V, it is possible to size and compare different energy storage solutions. Fig. 12 shows the volume distribution of the energy system for several alternatives for hydrogen storage and with oxygen stored as liquid. The battery stack is, in this case, very small and its volume is negligible when compared to the other parts. It is clear from the graph that the choice of solution for gas storage is essential. However, storing hydrogen with lower energy density, such as compressed gas at 200 bar, does not clearly favor the hybrid system over the full secondary battery system.

Fig. 13 compares the volume of three different energy systems as a function of the energy needs, i.e., as a function of the mission duration. In the hybrid system, liquid oxygen and liquid hydrogen are used. The simulations provide one intersection point: between the hybrid system and the rechargeable battery

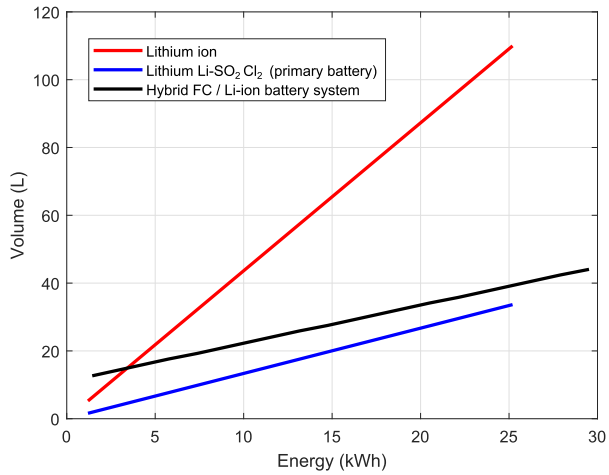


Fig. 13. Volume as a function of energy need for the hypothetical 20-h mission at 1260-W average power when using liquid hydrogen and oxygen storage for the hybrid system without neutral buoyancy.

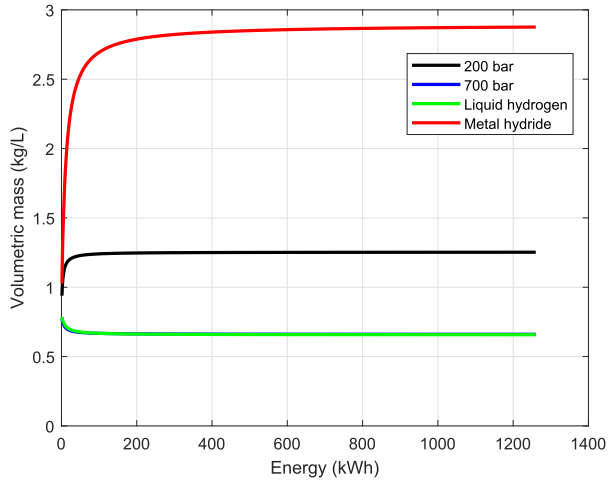


Fig. 14. Volumetric mass as a function of energy need for the extended hypothetical 1000-h mission for several hydrogen storage solutions when using liquid oxygen storage without neutral buoyancy.

system. This point occurs around 3 kWh of energy demand (see Fig. 13), and the difference in volume between these two system solutions keeps increasing with increasing energy demand. The volume of the primary battery system is the highest in these simulations, but the trend of the black (hybrid system) and blue (primary battery) curves is that they also will cross each other when more energy is needed, i.e., when longer missions are considered.

The designed mission is extended up to a 1000-h mission, leading to an energy consumption going from 0 to 1275 kWh. The volumetric mass of hybrid systems with different types of hydrogen storage and liquid oxygen, without the neutral buoyancy, is calculated and presented in Fig. 14. The 700-bar compressed hydrogen (blue) and the liquid hydrogen storage (green) follow exactly the same pattern; the volumetric mass first decreases rapidly when the energy need increases and then seems to reach an asymptote close to 0.66 kg/L. All four solutions reach an asymptote and both the 200 bar (black) and metal hydride (red) curves are above 1. Since volumetric mass of each system show an asymptotic behavior when the duration of the mission increases, the compensation needed will eventually be proportional to the time of the mission.

D. Generalization

It is interesting to compare the specific energy and the energy density of all the energy system solutions, as shown in Fig. 15, and for a long time simulation. To run this simulation, the hypothetical power profile was extended to up to a 1000-h mission. The total weight and volume were estimated for 1100, 500, and 1000 h. The hypothetical power profile can be compared to the short HUGIN mission in terms of t_1, t_2, P_{base}, and peak power demand. The primary battery shows good characteristics in terms of weight and volume, but it needs to be changed for each mission, which makes this solution less sustainable. Fig. 15 shows that when the hydrogen and the oxygen storage are well selected, the hybrid system's specific energy and energy density are higher than for the secondary battery system. The combination of liquid hydrogen and oxygen (yellow symbols) shows the best results in terms of specific energy but not in terms of energy density; indeed, the combination of liquid oxygen and metal hydride (green symbols) reaches a higher energy density. The combination of liquid oxygen and compressed hydrogen at 700 bar (purple symbols) also seems to be a promising solution since it follows the trend of the liquid oxygen and hydrogen combination.

Fig. 15 also shows how the specific energy and the energy density of the hybrid solution dramatically increase with the mission duration. In addition, when considering neutral buoyancy (squares), the same trend appears. The final specific energy and energy density are, therefore, not as high as without neutral buoyancy considered; however, they are still higher than the performance of the secondary battery. There is a clear influence of the compensation to reach neutral buoyancy on the final energy storage capacity of the different systems, especially for the metal hydride case. This trend further highlights the potential for using FC systems in AUVs. Finally, the simulation highlights an asymptote in the energy density and specific energy for a hybrid system when the duration becomes very long for all the combinations. This behavior was expected since when the energy storage increases, the FCs size and weight become negligible in the total system.

V. Discussion

The sizing strategy developed in this article shows a clear trend in favor of hybrid systems. However, some assumptions can be discussed. First of all, the different parameters of each energy storage solution and the overall efficiency of the battery and the FC system are considered. Additionally, the tubing part for the hybrid FC/battery system has not been considered in this study. For this reason, the real buoyancy is likely to change since free space is not considered when the volumetric mass is estimated and will, therefore, be lower than the calculated one. To consider the buoyancy as a major parameter has not been the scope of this study, but, of course, this might decrease the

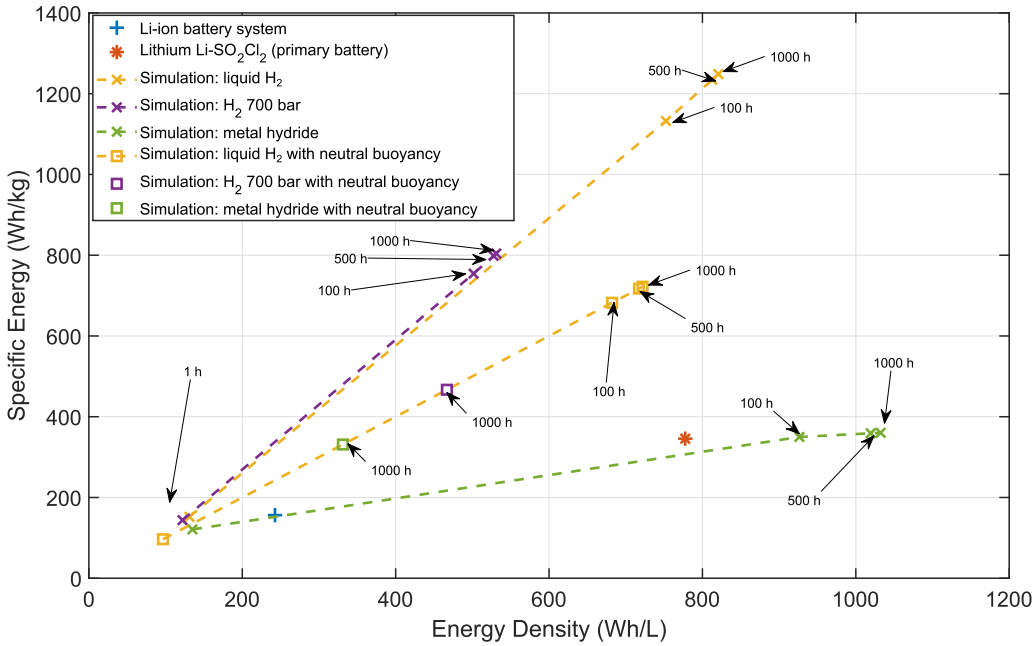


Fig. 15. Energy density and specific energy for the short HUGIN mission, the extended hypothetical power profile simulations, and the full battery solutions. For the hybrid system, oxygen is stored as liquid and several solutions for hydrogen storage are considered with an optimized FC output power without and with neutral buoyancy. The two battery systems correspond to the batteries presented in Table IV.

volume efficiency of the hybrid solution. The shifting point in Fig. 13 is expected to move; however, the general trend in the results shown in this article should be similar. An important part of future research should be to further investigate the scalability and energy density of complete systems.

Furthermore, it is important to put into perspective the good performance of the primary battery. This solution is worthwhile for a very long mission to achieve a specific goal, but it is not suitable for repeated missions since the battery needs to be changed for every mission. Primary batteries also typically have issues under high power demand, which can lead to overheating of the system [36]. The compensation needed to reach neutral buoyancy is also very important when considering the two battery cases.

Although not presented in Section IV, other oxygen storage systems (compressed at 153 bar and H_2O_2) can also be considered. They have lower performance in terms of gravimetric and volumetric density than the liquid solution, but they also show that when the mission becomes long and energy intense, they are volume and weight efficient.

Furthermore, as seen in Figs. 5, 9, and 12, when the mission duration increases and the FC output power is optimized, the battery pack does not increase that much in terms of size and the hybrid solution becomes very attractive compared to the full-battery solution. This also highlights the importance of optimizing the FCs output power, shown in Figs. 3, 7, and 11. If the FC is wrongly sized, the battery pack can become bulky and the benefits of the hybrid system will be reduced.

Even though it is important to reach neutral buoyancy in the final design, a low volumetric mass of the energy system is beneficial because this gives more design freedom. For example, the added weight can be used for trimming purposes, adding additional heavy sensors or more energy. On the other hand, when the energy system needs to be lighter, there is less design freedom in utilizing the volume to add payload capacity.

Moreover, the power split between the FC and the battery needs to be carefully considered in the future, since this analysis does not take into consideration any advanced energy management strategy including the AUV's dynamic behavior during a mission (speed, sensors ON/OFF, etc.). Many studies concerning energy management strategy for hybrid cars and buses have been performed in the last few years [11], [27], [28], [37]–[42] and it will be the next step for this project. In addition, the energy management strategy is crucial to increase the lifetime of the battery pack [43]. A dynamic model of the entire system will be created within the SMaRC project to simulate its behavior and highlight its capabilities. It is also very important to note that the flexibility in terms of total power output (i.e., speed of the AUV) is highly dependent on the battery since the FC is assumed to work at constant power. The optimization tends to minimize the battery size to the greatest extent possible, so it might be important to slightly oversize the battery pack to guarantee more freedom for the AUV.

In addition, when designing a real AUV, the ratio size/energy of the entire AUV can be interesting to estimate and can then be used to predesign underwater vehicles. The main idea is to determine the minimum size of a vehicle for a certain mission and endurance. Results show that a full secondary battery system for small AUV performing short-range missions is preferred, whereas an optimized hybrid system is the best solution for long-range AUVs.

Safety is also an important issue in regard to batteries and hydrogen, especially in a closed environment. Large batteries, both primary and secondary, could be hazardous.

Hydrogen can also be a problem and is highly explosive in some concentration ranges. Pressure around the energy storage hull can also become a problem for the vessel and all these safety issues will need to be carefully studied in the future to prove the practical relevance of a hybrid FC/battery system. This developed strategy does not investigate the difference between pressure-tolerant hybrid systems and enclosed atmospheric hybrid systems. Other practical aspects must be considered, for example, the condensation due to the FC has to be handled [44]. One solution is to include water reservoir in the system. These issues will be investigated in the future.

When building a real FC-powered AUV, heat management has to be considered given that the AUV may be working in cold environments, where both FC and battery need some heat to start and perform correctly. Hydrogen and oxygen tanks might also need to be maintained at certain temperatures. Since the AUV also produces heat (electric motor for propulsion, FC, sensors, etc.) a heat management strategy will be investigated in the future to ensure all the AUVs systems work properly. Studying the heat properties of the different hydrogen and oxygen storage solutions will also help the decision process. As with weight and volume, heat is a key parameter in selecting the best hydrogen and oxygen storage solutions.

Last but not least, even if the Ragone plot in Fig. 1 shows that an FC system has lower power density (watt per liter) than secondary batteries, the proposed combination of FC stack and battery pack always meet the power demand of the vehicle, as shown in Figs. 3, 6, and 10. In other words, such hybrid systems have potential to meet both the power and energy requirements of AUVs.

VI. Conclusion

In this article, a strategy for designing compact hybrid FC/battery systems for AUVs has been developed. Both real and user-designed power profiles are utilized to compare hybrid systems with secondary and primary battery systems in terms of weight and volume. Crucial aspects of a hybrid energy storage system for AUVs, i.e., FC stack, battery pack, battery SoC, hydrogen storage, oxygen storage, and compensation to reach neutral buoyancy, are considered. Results show that a well-designed hybrid system can be more compact in terms of stored energy per volume than the battery solutions and, consequently, increase the endurance of the vehicle.

The choices of fuel and oxygen storage for the FC are the main parameters influencing the final weight and volume of the vessel's energy system. Volume is the most limiting parameter in underwater applications. The optimization, in terms of volume, is a crucial procedure to ensure that the system can benefit from the advantages of the hybrid system. The optimized energy systems can meet the power requirement of the AUV at any time, and were tested for real power profiles.

The optimization shows promising results to select the best combination of FC stack, battery pack, and reactants storage to design the most compact hybrid system for AUVs. For long-range missions requiring larger amounts of energy, hybrid systems with well-selected hydrogen and oxygen storage solutions and a well-designed battery pack have the potential to be more volume efficient than the full-battery solution. For a small AUV performing short missions, rechargeable batteries are more suitable than the hybrid system, considering the higher complexity of the hybrid system and the small difference in terms of volume between these two solutions. Hybrid FC/battery power systems are more interesting than battery systems for AUVs when the mission duration increases. In addition, for long and cyclical missions, the optimization tends to give FC power close to the average power demand. However, when the mission is not very cyclical, there is a difference between the optimized FC power and the average power demand.

The strategy can be updated with the newest technology and used as a first step to estimate the relevance of FC implementation for several types of AUVs and underwater missions. Published data on FC powered AUVs show promising results. However, FCs are not yet found in commercial AUVs. The SMaRC project will provide further opportunities to analyze and implement this technology.

References

[1] A. Mendez, T. J. Leo, and M. A. Herreros, "Current state of technology of fuel cell power systems for autonomous underwater vehicles," *Energies*, vol. 7, no. 7, pp. 4676–4693, 2014.

[2] G. Schubak and D. Scott, "A techno-economic comparison of power systems for autonomous underwater vehicles," *IEEE J. Ocean. Eng.*, vol. 20, no. 1, pp. 94–100, Jan. 1995.

[3] Ø. Hasvold, N. J. Størkersen, S. Forseth, and T. Lian, "Power sources for autonomous underwater vehicles," *J. Power Sources*, vol. 162, no. 2, pp. 935–942, 2006.

[4] X. Wang, J. Shang, Z. Luo, L. Tang, X. Zhang, and J. Li, "Reviews of power systems and environmental energy conversion for unmanned underwater vehicles," *Renewable Sustain. Energy Rev.*, vol. 16, no. 4, pp. 1958–1970, 2012.

[5] A. M. Aris and B. Shabani, "An experimental study of a lithium ion cell operation at low temperature conditions," *Energy Procedia*, vol. 110, pp. 128–135, 2017.

[6] N.-C. Shih, B.-J. Weng, J.-Y. Lee, and Y.-C. Hsiao, "Development of a small fuel cell underwater vehicle," *Int. J. Hydrogen Energy*, vol. 38, no. 25, pp. 11138–11143, 2013.

[7] E. Raugel, V. Rigaud, and C. Lakeman, "Sea experiment of a survey AUV powered by a fuel cell system," in *Proc. IEEE/OES Auton. Underwater Veh.*, 2010, pp. 1–3.

[8] W. Hornfeld, "DeepC: The German AUV development project," STN ATLAS Elektronik GmbH, Bremen, Germany, Tech. Rep., 2002.

[9] Q. Cai, D. Brett, D. Browning, and N. Brandon, "A sizing-design methodology for hybrid fuel cell power systems and its application to an unmanned underwater vehicle," *J. Power Sources*, vol. 195, no. 19, pp. 6559–6569, 2010.

[10] P. Thounthong, V. Chunkag, P. Sethakul, B. Davat, and M. Hinaje, "Comparative study of fuel-cell vehicle hybridization with battery or supercapacitor storage device," *IEEE Trans. Veh. Technol.*, vol. 58, no. 8, pp. 3892–3904, Oct. 2009.

[11] A. De Risi, V. Paladini, T. Donateo, and D. Laforggia, "Control strategy optimization of a fuel-cell electric vehicle," in *Proc. ASME EFC05 Tech. Conf.*, 2005, pp. 14–16.

[12] D. Gao, Z. Jin, and Q. Lu, "Energy management strategy based on fuzzy logic for a fuel cell hybrid bus," *J. Power Sources*, vol. 185, no. 1, pp. 311–317, 2008.

[13] D. F. Waters and C. P. Cadou, "Estimating the neutrally buoyant energy density of a Rankine-cycle/fuel-cell underwater propulsion system," *J. Power Sources*, vol. 248, pp. 714–720, 2014.

[14] M. Winter and R. J. Brodd, "What are batteries, fuel cells, and supercapacitors?" *Chem. Rev.*, vol. 104, pp. 4245–4269, 2004.

[15] A. Chiche, C. Lagergren, G. Lindbergh, and I. Stenius, "Sizing the energy system on long-range AUV," in *Proc. IEEE/OES Auton. Underwater Veh. Workshop*, 2018, pp. 1–6.

[16] C. Spiegel, *PEM Fuel Cell Modeling and Simulation Using MATLAB*. Amsterdam, The Netherlands: Elsevier, 2011.

[17] R. d'Amore Domenech, M. A. Raso, A. Villalba-Herreros, Ó. Santiago, E. Navarro, and T. J. Leo, "Autonomous underwater vehicles powered by fuel cells: Design guidelines," *Ocean Eng.*, vol. 153, pp. 387–398, 2018.
[18] C.-H. Lee and J.-T. Yang, "Modeling of the Ballard-Mark-V proton exchange membrane fuel cell with power converters for applications in autonomous underwater vehicles," *J. Power Sources*, vol. 196, no. 8, pp. 3810–3823, 2011.
[19] I. Yamamoto *et al.*, "Fuel cell system of AUV 'URASHIMA'," in *Proc. MTS/IEEE Techno-OCEANS Conf.'*, vol. 3, 2004, pp. 1732–1737.
[20] M. Hitscherich, C. Cremers, D. Stolten, K. Pinkwart, and J. Tübke, "Development study of an air independent fuel cell system for an autonomous underwater vehicle (AUV)," *ECS Trans.*, vol. 75, no. 14, 2016, Art. no. 491.
[21] Y. Sone, M. Ueno, and S. Kuwajima, "Fuel cell development for space applications: Fuel cell system in a closed environment," *J. Power Sources*, vol. 137, no. 2, pp. 269–276, 2004.
[22] O. Siddiqui and I. Dincer, "A review on fuel cell-based locomotive powering options for sustainable transportation," *Arabian J. Sci. Eng.*, vol. 44, no. 2, pp. 677–693, 2019.
[23] M. Nadal and F. Barbir, "Development of a hybrid fuel cell/battery powered electric vehicle," *Int. J. Hydrogen Energy*, vol. 21, no. 6, pp. 497–505, 1996.
[24] H. S. Das, C. W. Tan, and A. Yatim, "Fuel cell hybrid electric vehicles: A review on power conditioning units and topologies," *Renewable Sustain. Energy Rev.*, vol. 76, pp. 268–291, 2017.
[25] T. Hyakudome, T. Tani, H. Ito, and K. Sugihara, "Concept of long cruising range autonomous underwater vehicle powered by fuel cell," in *Proc. MTS/IEEE OCEANS Conf.* Kobe, Japan, 2018, pp. 1–5.
[26] A. Rousseau, P. Sharer, and R. Ahluwalia, "Energy storage requirements for fuel cell vehicles," SAE, Warrendale, PA, USA, Tech. Rep. 2004-01-1302, 2004.
[27] P. Bubna, D. Brunner, S. G. Advani, and A. K. Prasad, "Prediction-based optimal power management in a fuel cell/battery plug-in hybrid vehicle," *J. Power Sources*, vol. 195, no. 19, pp. 6699–6708, 2010.
[28] T. Fletcher, R. Thring, and M. Watkinson, "An energy management strategy to concurrently optimise fuel consumption & PEM fuel cell lifetime in a hybrid vehicle," *Int. J. Hydrogen Energy*, vol. 41, no. 46, pp. 21503–21515, 2016.
[29] A. S. Mussa, M. Klett, M. Behm, G. Lindbergh, and R. W. Lindström, "Fast-charging to a partial state of charge in lithium-ion batteries: A comparative ageing study," *J. Energy Storage*, vol. 13, pp. 325–333, 2017.
[30] S. Saxena, Y. Xing, D. Kwon, and M. Pecht, "Accelerated degradation model for C-rate loading of lithium-ion batteries," *Int. J. Elect. Power Energy Syst.*, vol. 107, pp. 438–445, 2019.
[31] F. Benavente-Araoz, M. Varini, A. Lundblad, S. Cabrera, and G. Lindbergh, "Effect of partial cycling of NCA/graphite cylindrical cells in different SoC intervals," *J. Electrochem. Soc.*, vol. 167, no. 4, 2020, Art. no. 040529.
[32] Horizonfuelcell. Accessed: Jan. 2018. [Online]. Available: https://www.horizonfuelcell.com/h-series-stacks.
[33] Powercell. Accessed: Jan. 2018. [Online]. Available: https://www.powercell.se/en/products-and-services/fuel-cell-stacks/
[34] K. L. Davies and R. M. Moore, "UUV FCEPS technology assessment and design process," Hawaii Natural Energy Inst. (HNEI), School Ocean Earth Sci. Technol. (SOEST), Univ. Hawaii Manoa, HI, USA, Tech. Rep. 061027, 2006.
[35] em3ev. Accessed: Feb. 2018. [Online]. Available: https://em3ev.com/product-category/ebike-battery-parts/ebike-battery-in-case/
[36] T. Reddy and D. Linden, *Linden's Handbook of Batteries*. New York, NY, USA: McGraw-Hill, 2010.
[37] K. Ettihir, L. Boulon, and K. Agbossou, "Optimization-based energy management strategy for a fuel cell/battery hybrid power system," *Appl. Energy*, vol. 163, pp. 142–153, 2016.
[38] L. Xu, J. Li, J. Hua, X. Li, and M. Ouyang, "Adaptive supervisory control strategy of a fuel cell/battery-powered city bus," *J. Power Sources*, vol. 194, no. 1, pp. 360–368, 2009.
[39] G. Li, J. Zhang, and H. He, "Battery SoC constraint comparison for predictive energy management of plug-in hybrid electric bus," *Appl. Energy*, vol. 194, pp. 578–587, 2017.
[40] Z. Hong, Q. Li, Y. Han, W. Shang, Y. Zhu, and W. Chen, "An energy management strategy based on dynamic power factor for fuel cell/battery hybrid locomotive," *Int. J. Hydrogen Energy*, vol. 43, no. 6, pp. 3261–3272, 2018.
[41] J. Han, J.-F. Charpentier, and T. Tang, "An energy management system of a fuel cell/battery hybrid boat," *Energies*, vol. 7, no. 5, pp. 2799–2820, 2014.
[42] J. Garus and J. Małecki, "Simulation model of fuel cell energy systems for AUV," in *Proc. IEEE 27th Conv. Elect. Electron. Eng.*, 2012, pp. 1–4.
[43] E. Schaltz, A. Khaligh, and P. O. Rasmussen, "Influence of battery/ultracapacitor energy-storage sizing on battery lifetime in a fuel cell hybrid electric vehicle," *IEEE Trans. Veh. Technol.*, vol. 58, no. 8, pp. 3882–3891, Oct. 2009.
[44] H. Weydahl, M. Gilljam, T. Lian, T. C. Johannessen, S. I. Holm, and J. Ø. Hasvold, "Fuel cell systems for long-endurance autonomous underwater vehicles — challenges and benefits," *Int. J. Hydrogen Energy*, vol. 45, no. 8, pp. 5543–5553, 2020.



Ariel Chiche was born in Paris, France, in 1994. He received the M.Sc. degree in engineering in physics, electronics, and materials sciences in 2017 from Grenoble INP – PHELMA, Grenoble, France, where he specialized in electrochemistry and processes for the energy and the environment. He is currently working toward the Ph.D. degree with the Division of Applied Electrochemistry, KTH Royal Institute of Technology, Stockholm, Sweden.

His work is on the implementation of fuel cell systems for underwater and maritime applications for which he investigates both the design of such systems and the behavior of fuel cells in such an environment.



Göran Lindbergh received the M.Sc. and Ph.D. degrees in 1985 and 1991, respectively.

Since 2003, he has been a Professor of Chemical Engineering with the KTH Royal Institute of Technology, Stockholm, Sweden. He is heading the Applied Electrochemistry group working with electrochemical power sources and electrolytic processes within the field of electrochemical engineering. A common theme in the ongoing research projects is the mathematical modeling and electrochemical characterization of electrochemical systems.



Ivan Stenius received the M.Sc. degree in lightweight structures from the KTH Royal Institute of Technology (KTH), Stockholm, Sweden, in 2003, and the Doctor in Technology degree in lightweight structures on hydroelasticity and fluid structure interactions on high-speed craft from KTH in 2009.

He is currently a full-time Associate Professor with the Department of Aeronautics and Vehicle Engineering, KTH. He has expertise in underwater vehicle modeling design and construction, as well as maneuvering and navigation. He has been the lead developer of design and analysis software tools in collaboration with FMV and The Swedish Coast Guard. In the recent years, he has been heavily involved in building up a research group in underwater technology and maritime robotics with KTH and is currently a PI of the Swedish Maritime Robotics Centre hosted by KTH that involves research on underwater robotics.



Carina Lagergren received the Ph.D. degree in 1994.

She is currently a Professor with the Department of Chemical Engineering, KTH Royal Institute of Technology, Stockholm, Sweden. Her research focus is in electrochemical systems and has mainly been within the fuel cell area, in which both high-temperature molten carbonate fuel cells and polymer electrolyte fuel cells, acid as well as alkaline, have been studied. Her research interest also includes electrolysis of, e.g., water at high temperature for production of fuels.